Google Cloud



Build Software Sustainably

AI that powers a resilient future

Table of contents



Part 1:
Understanding the impact of AI 04

Part 2:
Building on a Sustainable by Design AI infrastructure 09

Part 3:
The road ahead: AI for a more resilient future 29



Foreword

At Google, we strongly believe innovation and sustainability go hand in hand. AI is reshaping our world, giving customers the opportunity to measure, optimise, grow and build solutions at scale. This progress brings a pressing need to define the roles of both the cloud providers and software developers, highlighting how choosing the right infrastructure is critical for building sustainably.

We're building the world's most energy-efficient computing infrastructure, advancing water stewardship, and strengthening energy grids. This infrastructure is the foundation for sustainable transformation with AI.

This ebook puts ownership into action. It's for every technologist, product manager, designer, engineer, and data professional aiming to build efficient software that not only reduces environmental impact but also cuts costs, both of which are critical for building resilience.

However, we need to be clear-eyed about AI's footprint. Developing and running AI requires energy and resources, but it also creates immense opportunities for positive sustainability and financial impact. Here is what that looks like in action:

Green Light: In partnership with Google Maps, we're optimising traffic lights which can help reduce vehicle emissions in cities. Green Light, an AI-based tool that helps city traffic engineers optimise the timing of traffic lights, has shown the potential to reduce stops at intersections by up to 30% and reduce emissions at intersections by an average of over 10%[a].

Project Contrails: Condensation trails (contrails) are clouds created by airplanes flying through humid areas. These account for roughly a third of aviation's global warming impact. In one trial consisting of 70 test flights, flights that adjusted their routes using our AI-based contrail predictions showed a 54% reduction in contrail kilometers, when compared to control flights that didn't have access to the AI predictions[b].

Solar API: As part of Google's Environmental APIs, the Solar API provides detailed rooftop data to help installers, businesses and public authorities estimate the potential of a solar installation and prepare the planning.

WeatherNext: This family of AI models from Google DeepMind and Google Research produces state-of-the-art weather forecasts. These are more efficient than traditional models, helping save lives and enhance the reliability of energy and supply chains.

Part 1

Understanding what's needed to run AI

01 Foundations of sustainable software and AI

AI is a game changer. But like any powerful tool, what matters is how it's used. Understanding how cloud providers are working to reduce energy consumption, water use and CO2 emissions is the first step to building smarter solutions.

There are three core concepts that are essential to understand when looking at the sustainability of software development. Organisations such as the Green Software Foundation (GSF) offer excellent direction and frameworks but let's start with the three core concepts:

1

Carbon efficiency

Carbon efficiency means optimising an application's entire footprint, from development to usage emissions. To build software sustainably at scale, it is crucial to optimise these workloads.

2

Energy efficiency

Energy efficiency is about performing functions with minimal energy consumption. In many cases, the primary emissions contribution comes from electricity consumption of AI hardware, highlighting the importance of designing accelerators that achieve more with less energy.

3

Carbon intensity

Carbon intensity refers to the varying carbon levels in an electricity grid based on the regional energy mix. It's impacted by factors like weather and demand and is typically measured in gCO2eq per kWh.

02 Energy and resource consumption

Understanding AI's environmental impact requires a holistic, life-cycle perspective. It extends beyond energy consumption and includes the entire infrastructure, from manufacturing of hardware to operating data centers and resources, like water, needed to keep them running.

Energy

Powering AI requires electricity, and we recommend that developers consider the carbon impact of their applications. At Google, we report our energy consumption and we empower customers to leverage our 24/7 carbon-free energy efforts and consider the carbon impact of where they locate their applications. Learn more about carbon free energy for Google Cloud regions.

Training

AI model training, particularly for large language models (LLMs), requires substantial energy, often running on thousands of specialised accelerators (GPUs or TPUs) for days at a time. This initial energy use and the hardware's embodied carbon must be fully accounted for.

Inference

AI inference is the process of using a trained machine learning model to make predictions or decisions based on incoming user data. It is the phase where the model applies what it learned during training to solve problems. Because of this large-scale deployment, it is important to establish transparent and comprehensive methodologies for calculating the energy demand of inference.

Many AI energy consumption calculations only include active machine consumption. As a result, they represent theoretical efficiency instead of true operating efficiency at scale. A truly comprehensive calculation must account for all critical elements of serving AI globally and should include idling compute power with all involved hardware elements, not only the energy required for AI accelerators.

Embodied Carbon

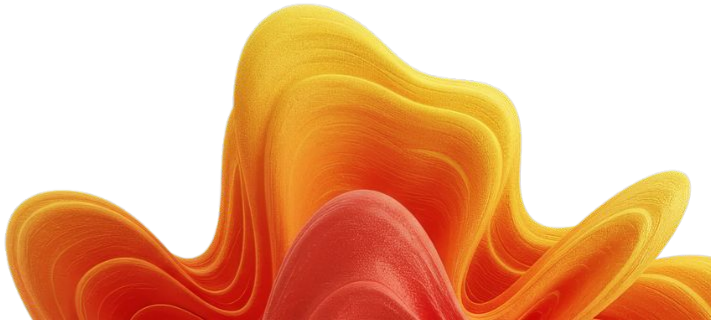
A complete assessment of AI's footprint must include the "embodied carbon" from the entire lifecycle of hardware, from manufacturing to disposal. For our own AI accelerator hardware, the Tensor Processing Units (TPUs) we are constantly eager to better understand and reduce these emissions and we pioneered a comprehensive Life-Cycle Assessment (LCA) for our AI hardware.

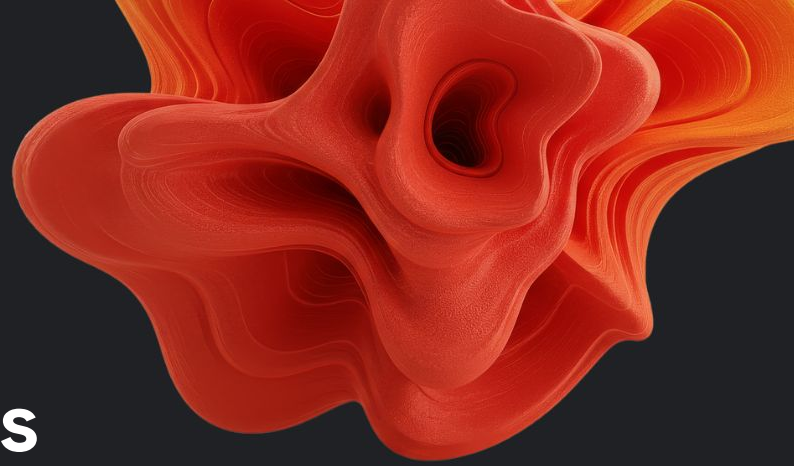
Data Centre Efficiency

The efficiency of data centre infrastructure is critical; our fleet-wide average Power Usage Effectiveness (PUE) of 1.09 makes our data centres among the industry's most efficient[d].

Water Usage

Beyond compute, data centre overhead includes the water needed for cooling systems, which are often more energy-efficient than air cooling. We are committed to responsible water stewardship, with a goal to replenish 120% of the freshwater volume we consume, on average, by 2030 through global restoration and conservation projects.





Understanding AI's resources helps developers make more responsible decisions from the start.

For a deeper dive into AI hardware's emissions, explore our study on the Measuring the Environmental Impact of AI Inference and and our Life-Cycle Emissions of AI Hardware.

Part 2

Building on a Sustainable by Design AI infrastructure

This section provides actionable guidance for everyone in the software lifecycle, from product managers to developers, on how to build more sustainable AI-powered software.

Introducing “Sustainable by Design”

AI architecture that follows Google’s 4Ms best practices.

1. Using **purpose-built AI processors**
2. Selecting **efficient models** fit for purpose
3. Automating **data and analytics at scale** with zero-ops services
4. Hosting workloads in **cleaner energy locations** where possible



01 The 4Ms: Machine, Model, Mechanisation, and Map

Machine

Select the right machine for your AI workload. Google Cloud offers TPU, GPU, and CPU options optimised for performance and energy efficiency. For example, the first Google TPU designed to power thinking, inferential AI models at scale, Ironwood, is nearly 30x more power-efficient than our first Cloud TPU from 2018[e]. This choice helps maximise workload performance while minimising energy consumption, costs and carbon impact.

Model

Application builders should build or select accurate, efficient models that use a fraction of the resources. An example is the Earth AI model family purpose-built for highly specialised sustainability use cases in collaboration with DeepMind, Research, and Geo.

Mechanisation

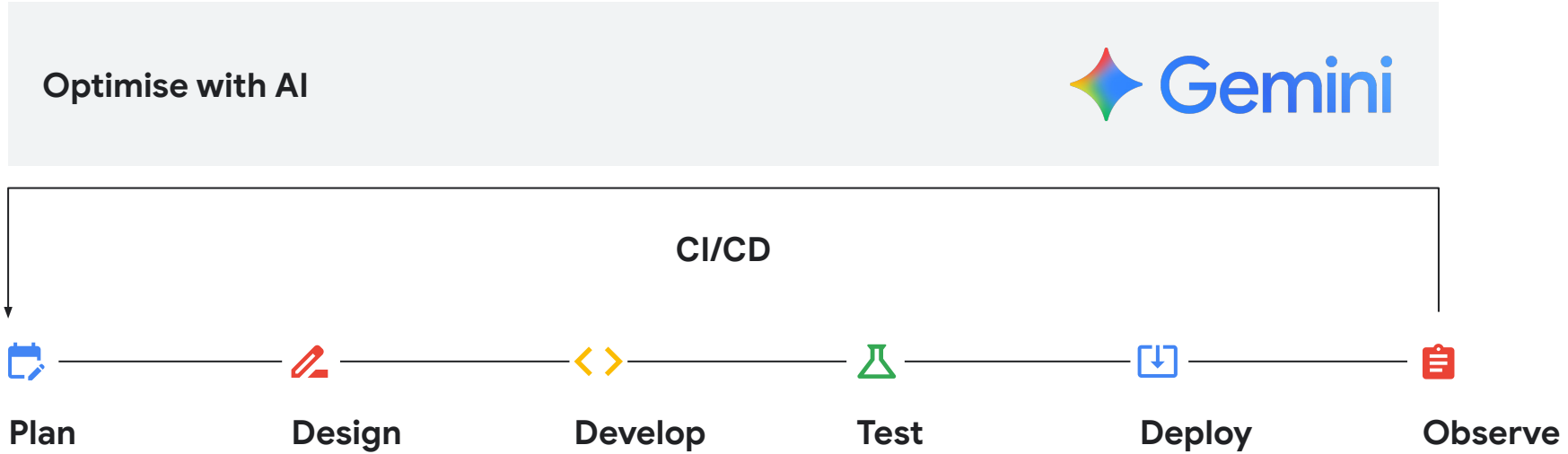
Use intelligent, serverless data, analytics, and AI services like Vertex AI, AI Hypercomputer, BigQuery, and Spanner to handle massive data volumes with superior efficiency. The cloud provider manages the underlying infrastructure, for example, through intelligent auto-scaling to help you achieve greater operational and carbon efficiency.

Map

Choose where to host your AI workloads based on regional energy profiles. Google Cloud offers tools to help you optimise for price, performance, and environmental outcomes by selecting regions with high percentages of carbon-free energy.

02 AI in the software development cycle

AI is reshaping the traditional Software Development Life Cycle (SDLC) and offers significant opportunities to incorporate sustainability. Here's how.



Plan: Set the starting point

Planning is the phase where you define the scope and purpose of your software. Typically handled by product managers and designers.

01 Define sustainability targets

Start by setting specific targets for products or features, like total application emissions or GPU hours per user request.

02 Incorporate into product management process

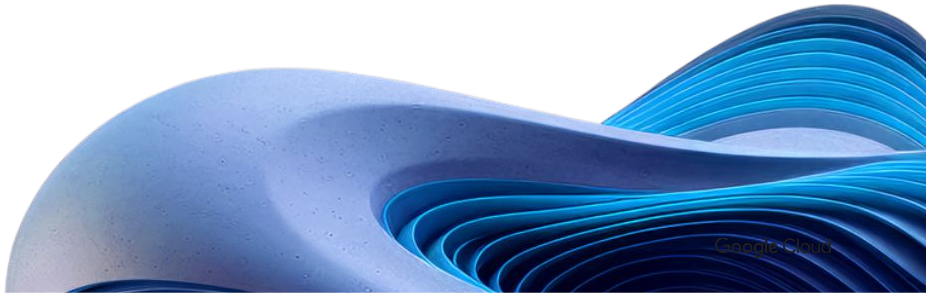
Integrate sustainability KPIs into your process so that new products and features are also selected based on their sustainability impact.

03 Distribute ownership

Drive sustainability through your organisation by defining ownership and setting efficiency Objectives & Key Results (OKRs) for all teams.

04 Be transparent

Educate users on your product's environmental impact, for example, by showing the cost savings and fuel savings of a fuel-efficient route.

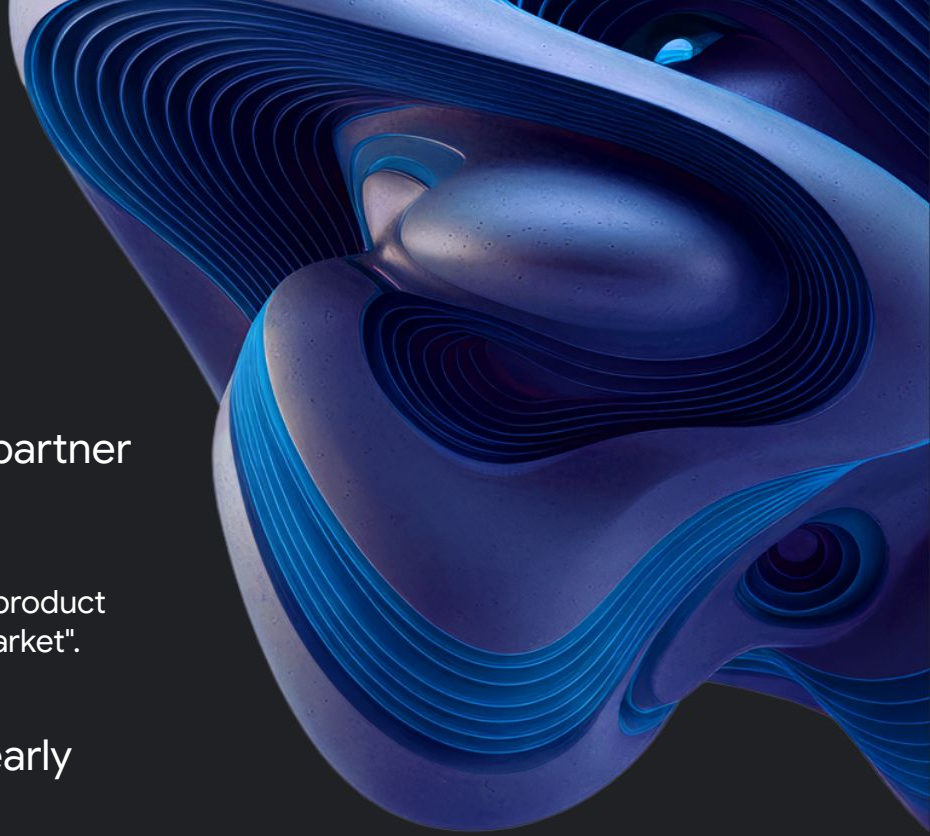


How AI can help you.

Use AI assistants like Gemini as a sparring partner for brainstorming and generating ideas.

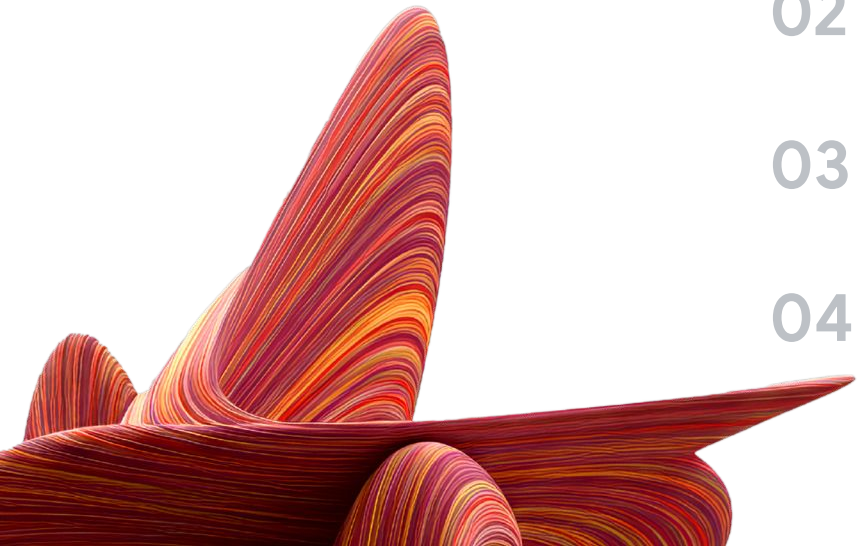
> For example, "Analyse how we could differentiate our product with additional sustainability features in our current market".

Tools like Stitch can also assist in creating early prototypes for user validation.



Design:
Define an efficient architecture & platform

In the design phase, CTOs and software architects can shape an application's design and underlying structure:



01 Choose the platform

When selecting a cloud platform, ensure it allows you to measure your footprint. Gathering efficiency details and being able to track metrics like kWh or CO2 emissions provides a measurable foundation for your design.

02 Choose higher level services

Using managed services often improves resource efficiency. For example, a serverless platform like Cloud Run means Google handles the efficient scheduling of resources, allowing services to scale from zero to thousands of instances and back down automatically.

03 Design scalable architectures

A major opportunity for saving resources is designing applications that scale elastically with user load, using resources only when requested and scaling down rapidly when not needed. Modern microservices and serverless runtimes provide a great foundation for these architectures.

04 Create change-driven data pipelines

Invest in making data processing architectures change-driven and event-based. An outdated batch-focused data pipeline leads to poor resource usage, whereas an event-based approach ensures only new data is processed, optimising resources efficiently.



How AI can help you.

AI can act as your reviewer, offering feedback on architectures and validating their efficiency. Sketch a rough architecture and submit it to Gemini for feedback, specifically requesting analysis of carbon efficiency, energy proportionality or alignment with the 4Ms and sustainable software principles. Alternatively, use Google Cloud Assist (a side panel in Cloud Console) to propose sustainable architectural designs.

Develop:
Build smarter, engineer for resilience

This is the engineering phase, where developers bring product ideas and defined architectures to life. AI increasingly supports this process through code generation and completion, helping you embed sustainable coding practices.

01 Right programming language

Evaluate and choose the appropriate programming language for the task, as each has unique strengths and resource profiles. For example, Golang is great for high-concurrency with low resource overhead, while Python offers extensive libraries and the flexibility to leverage accelerators for data-intensive operations.

02 Code efficiency and performance

Lean, streamlined code directly contributes to a smaller carbon footprint. Focus on reducing overhead, eliminating unused code, and ensuring overall code efficiency. Applying the Pareto principle: concentrate optimisation on the 20% of code with the highest execution frequency.

03 Resource usage pattern

Beyond efficient code, it's crucial to optimise the resource usage of individual services. A service's resource use should grow in proportion to its amount of requests. If demand doubles, its CPU and memory usage shouldn't more than double, or it will become inefficient at scale.



How AI can help you?

AI tools, like Gemini CLI or Gemini Code Assist, can help speed up development through code generation. When using them, provide instructions to focus on efficient code. This guides the AI toward sustainable solutions and allows it to learn from your existing, well-optimised code.



Testing's evolution: CI/CD pipelines that spark innovation

Testing ensures product quality and sits at the intersection of development and deployment. Over the decades, testing has expanded, consuming more resources for CI/CD pipelines that test updated code in realistic environments. Engineers and DevOps experts collaborate to limit the resource demands of these systems.

01 Efficiency & caching

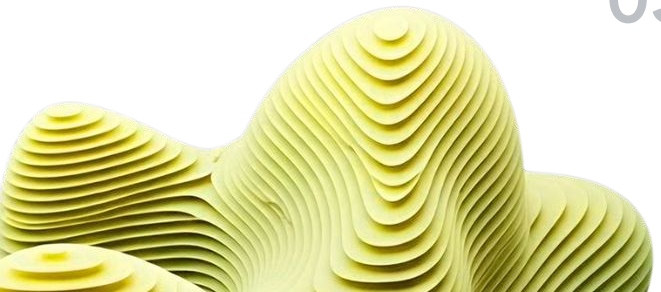
Invest in your CI/CD pipeline's efficiency by optimising or reducing resource-intense steps. Given their high frequency, even small optimisations yield significant benefits. Incorporate effective caching, such as Kaniko for container build pipelines, to reduce resource needs and accelerate cycles.

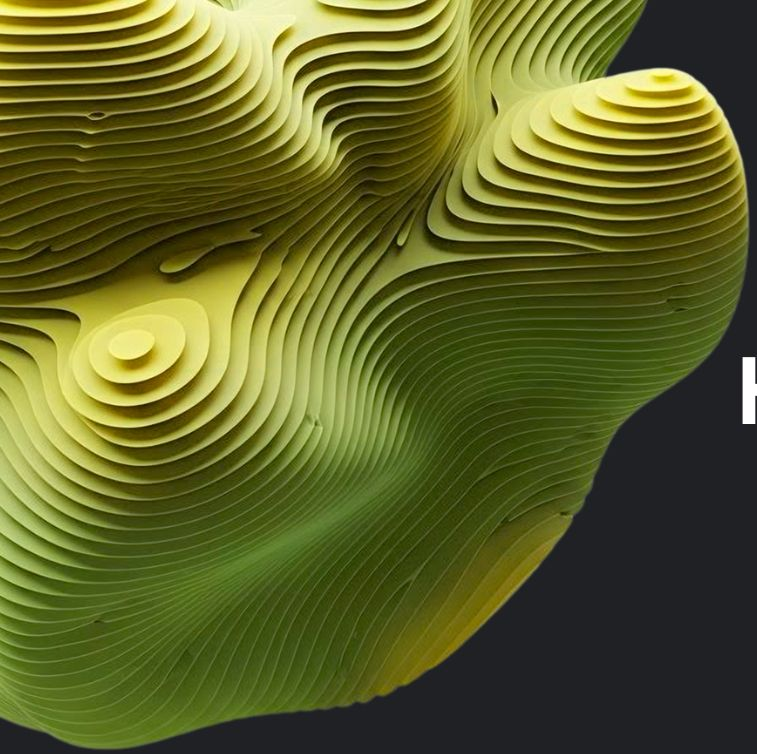
02 Reduce testing cycles

As testing and deployment pipelines become more efficient, teams tend to run them more often, a classic case of efficiency increasing demand (sometimes known as the Jevons paradox). While modern development processes, like trunk-based development drive this, teams can still reduce test cycles. Reflect on your processes and find ways to either minimise tested code updates or separate checks into continuous and less frequent runs.

03 Test & measure resource usage

Measurement is foundational for any optimisation. CI/CD pipelines are an excellent starting point. Mechanisms like Google Kubernetes Engine's (GKE) usage metering per namespace allow you to measure every build, integration, or test cycle. This provides valuable data on resource usage (CPU, Memory) for a long term trend analysis.





How AI can help you.

AI is transforming development processes, impacting testing by enabling quicker submissions and more frequent integrations. AI agents can perform early checks, identify potential errors, and side effects between parallel pushes, and even help redesign development for less resource-intensive testing. AI will help analyse current pipelines and propose optimisations for continuous, high-impact resource reduction.

Deploy for impact: Efficiency at scale

In this stage, previously optimised software is deployed to production. This section focuses on backend software operated by DevOps experts. While earlier phases address embodied carbon, this stage generates runtime carbon emissions. Even the smallest piece of overlooked code can run billions of times, creating significant impact. This makes best practices for efficient operation vital.

01 Leverage low-carbon regions & times

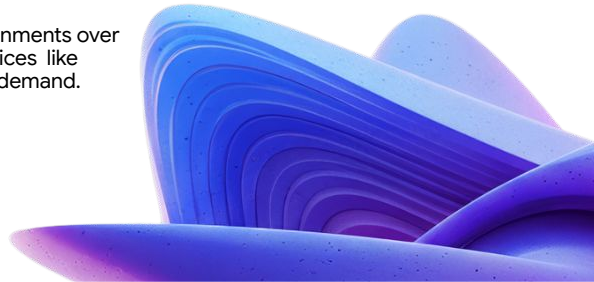
The energy mix varies significantly by deployment location. Google Cloud provides users with carbon-free energy data, enabling informed decisions for regions with high percentages of renewable energy. Organisations can set policies that make low-carbon regions the default, requiring teams to justify any exceptions. Alternatively, you can optimise workloads on time. Using data on the hourly energy mix of a grid, you could schedule workloads for times when carbon intensity is the lowest.

02 Rightsize

Whether serverless or VM-based, rightsizing is crucial to avoid wasting compute resources. Understand your software's resource needs and scaling behaviour to ensure it runs in an environment with the right resource mix. Custom machine types suit VMs, while serverless platforms require you to specify requested resources, such as in a Kubernetes deployment. Horizontal scaling (adding instances) is typically more efficient than vertical scaling (more resources per instance), complementing microservices. For AI models, rightsizing means understanding accelerator (GPU/TPU) needs for inference, and GPU time-sharing can efficiently manage accelerator access.

03 Remove idle workloads

This practice is simple but often overlooked. Software systems not required at certain times, like non-productive environments over a weekend, should not occupy active compute resources. Ensure your platform allows you to stop them, or run on services like Cloud Run, which automatically distributes workloads, minimises idling, scales down to zero, and can request GPUs on demand.



How AI can help you.

For years, AI models in Google Cloud console have recommended deployment improvements like rightsizing and identifying idling VMs or unused projects. The new Google Cloud Assist acts as a personal DevOps assistant directly in Cloud Console, simplifying operations and pinpointing inefficiencies.

> For example, ask Gemini Cloud Assist, “Based on the resource utilisation of VM X, please identify times with a heavy under-utilisation".

You can also enable the Carbon Footprint tool in your Google Cloud console to measure, report and reduce your organisation’s IT emissions.

Observe for insights: Monitor, identify and repeat

Optimisation begins with measurement. In this phase, keep these tips in mind:

01 Leverage Cloud Carbon Footprint

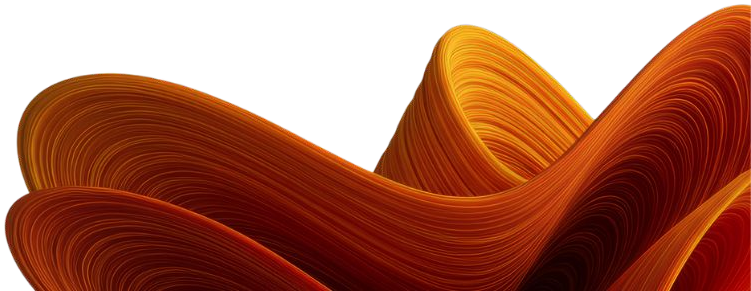
The carbon footprint report provides essential data on your Google Cloud carbon emissions, detailed by project, product, and location. Enhance this by linking carbon data with billing export details in BigQuery for deeper analysis.

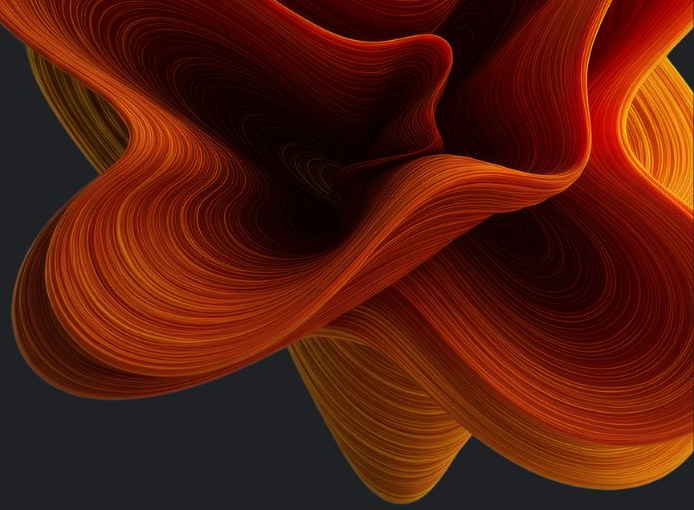
02 Review code executions

Establish mechanisms, to log metrics on execution, such as counts and times for specific sections. This data is valuable for pinpointing areas with the greatest potential for efficiency gains and guiding ongoing optimisation efforts.

03 Set sustainability budgets or targets

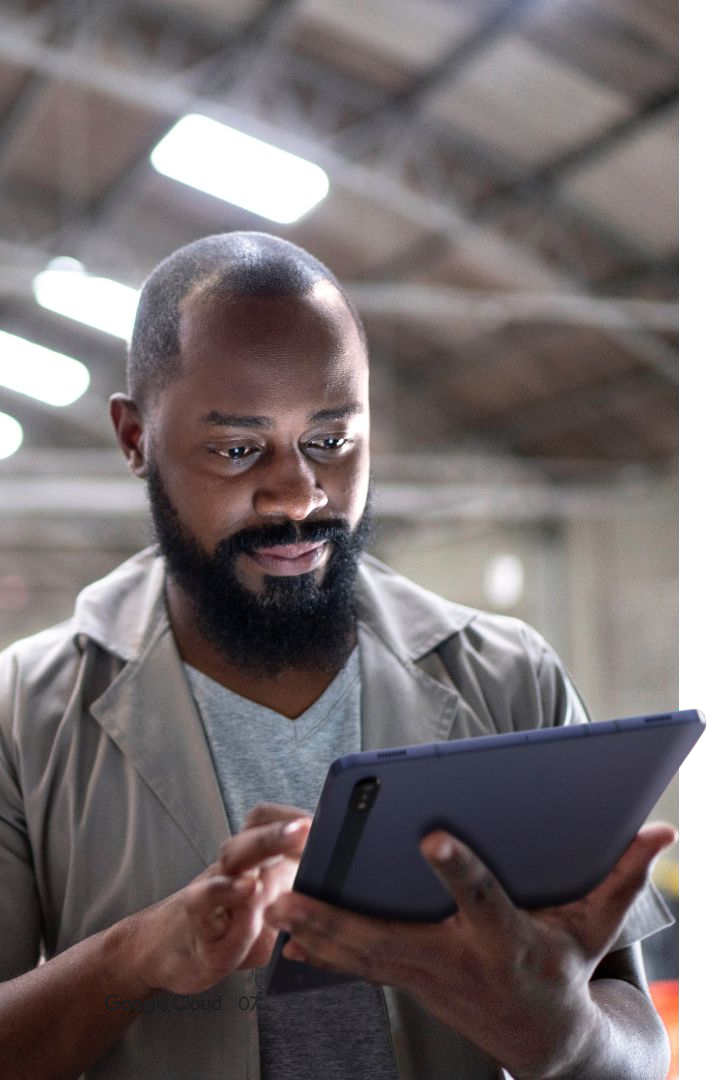
Adopt a practice from Site Reliability Engineering (SRE) by defining sustainability targets or budgets, similar to error budgets. For example, you could specify that a project should not exceed a certain kgCO2eq.





How AI can help you.

The more data you measure, the more AI can assist in detecting patterns and providing actionable insights. Gemini in BigQuery is an effective tool for analysis. Alternatively, ask Gemini Cloud Assist to investigate logs and pinpoint improvements.

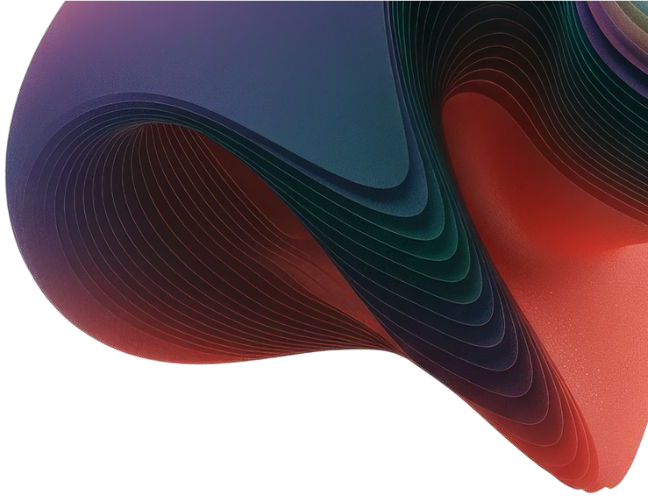


03 It's about the models: AI that builds and deploys efficiently

The previous section covered making the core development process more sustainable with AI's help. This section focuses on AI training and models themselves. When building or running your own models, keep the following in mind.

1. Designing for a lighter footprint
2. Smarter deployment and operations
3. Reuse, don't rebuild

Designing for a lighter footprint



Apply powerful optimisation techniques during development to significantly reduce a model's size and computational requirements.

Pruning

Apply common pruning techniques to remove unnecessary weights (nodes and synapses) from a trained model. This reduces its complexity and computational demands without significantly impacting performance.

Post-training quantisation

This technique makes a model more efficient by converting its weights into simpler data types. This reduces memory usage and speeds up calculations.

Model distillation

This process is like an expert teaching a student. Train a smaller, more efficient 'student' model to copy the performance of a larger 'teacher' model, capturing its knowledge in a more compact and energy-efficient form.

Smarter deployment and operations

How and where you deploy your model significantly influences its energy consumption.

Pick your region and your timing

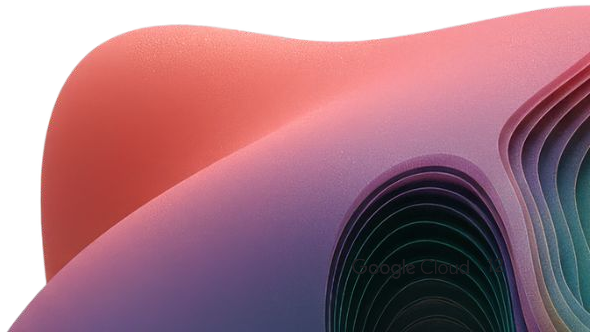
AI training often requires large datasets, so training runs should be located close to the data. Carefully select this location based on regional carbon efficiency, as training jobs are typically portable and data-reliant.

Leverage efficient hardware

Utilise AI accelerators, like Google TPUs or the latest GPUs to maximise performance per watt. This contributes to more sustainable model operations. It's also crucial to minimise idling for these accelerators.

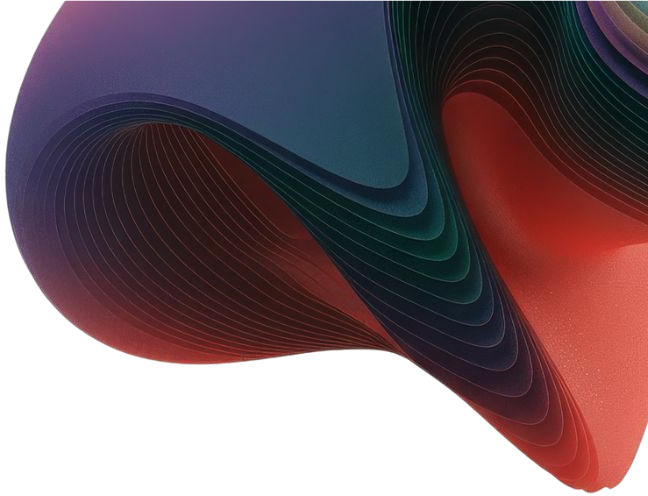
Track your experiments

Use tools and mechanisms to meticulously track training runs. This helps avoid redundant work and identify the most resource-efficient configurations for future model development and deployment.



Reuse, don’t rebuild

Reusing existing work is essential when developing new models. It optimises research capacity and expert resources, and also saves time and computational resources.



Leverage existing models to retrain

Check for open source or licensable models that meet your needs and allow fine-tuning. This significantly reduces resource demands for training. Large language models (LLMs) and multimodal models like Gemini now solve specific problems that previously required specialised models, such as visual inspection and document extraction. Alternatively, the open-source Gemma models allow you to further adapt, tune and custom-deploy them.

Leverage existing architectures

If no existing model fits, look for proven, effective architectures for classical ML models. Many choices are available, like MobileNet or SpineNet, which can streamline your development process.

Part 3

The road ahead: AI for a more resilient future

AI, when used right, can help organisations reduce their environmental impact and drive resilience. That's why 'sustainable AI' practices are crucial. This eBook shows you how to build with sustainability in mind. Now, let's make it happen.

Your turn to build software sustainably

This is not just theory. Every decision, from architecture to code, plays a part. Here is how you and your organisation can get started:

Choose the right Cloud Provider: Why Google Cloud?

- Our data centres are among the industry's most efficient, operating at a fleet-wide average Power Usage Effectiveness (PUE) of 1.09 in 2024. A PUE of 1.0 is the ideal, indicating perfect efficiency. This is a metric used to measure the energy efficiency of a data centre, calculated by dividing the Total Facility Energy by the energy consumed solely by the IT Equipment. The resulting ratio indicates how much of the energy is used for overhead functions like cooling and lighting.
- Google's seventh-generation Tensor Processing Unit (TPU), Ironwood, is nearly 30 times more power efficient than our first Cloud TPU from 2018[e].
- Our serving stack makes highly efficient use of CPUs and minimises TPU idling by dynamically moving models based on demand in near real-time.
- Our software efficiency efforts and clean energy procurement have driven a 33x reduction in energy consumption and a 44x reduction in the carbon footprint for the median Gemini Apps text prompt over a recent 12 month period[f].
- Using our comprehensive methodology in May 2025, we estimated that a median Gemini Apps text prompt consumes 0.24 Wh of energy, less than watching nine seconds of television and substantially lower than many public estimates[f].
- Despite continued growth of electricity consumption from 2023 to 2024, Google's Scope 2 MB emissions decreased over the same timeframe. This demonstrates an important decoupling between electricity consumption and emissions impact for Google's data centres. And in 2024, we achieved at least 80% carbon-free energy on an hourly basis across 9 of 20 grid regions with Google-owned-and-operated data centres.

Build on a data foundation

Data is key for any optimisation and improvement. You can only optimise what you can measure. Establish robust measurement of your resource usage and environmental footprint to understand your impact and track progress.

Get started

The most important thing in this journey is to get started somewhere. Reimagine your software products, services and processes for minimal environmental impact, applying the practices that you have learned here.



Together, we can reduce tech's environmental impact and unlock new solutions. Google Cloud is committed to providing the tools, platforms, and partnership to support you on this journey.

Glossary

AI and sustainability terms, demystified

AI (Artificial Intelligence)
The theory and development of computer systems able to perform tasks that normally require human intelligence.

Carbon awareness
The practice of scheduling or shifting computational workloads to times or locations where electricity has a lower carbon intensity (i.e., is cleaner).

Carbon efficiency
A principle of green software focused on emitting the least amount of carbon possible for a given task.

Carbon Free Energy Percentage (CFE%)
Average percentage of carbon free energy consumed in a particular location (e.g. Google data centre) on an hourly basis, while taking into account the investments we have made in carbon-free energy in that location.

CI/CD (Continuous Integration/Continuous Deployment/Delivery)
Practices that automate the building, testing, and deployment of software.

Embodied carbon/emissions
The greenhouse gas emissions associated with the manufacturing, transportation, and end-of-life disposal of hardware or physical products.

GreenOps
The practice of bringing financial accountability and environmental sustainability to cloud operations.

GPU (Graphics Processing Unit)
GPUs can be used to accelerate machine learning (ML), data processing, and graphics-intensive workloads on your virtual machine (VM) instances

Location Based Emissions
Refers to the measure of the carbon emissions from an entity's electricity consumption by applying the average carbon intensity of the local power grid to that consumption, regardless of specific energy contracts.

Green Software Foundation (GSF)
A non-profit organisation focused on building a trusted ecosystem of people, standards, tooling, and best practices for green software.

Inference (AI)
The process of using a trained AI model to make a prediction or generate an output based on new input data.

Distillation
A model compression technique where a smaller "student" model is trained to mimic a larger, pre-trained "teacher" model.

Pruning (AI Model)
A technique to remove redundant or less important parameters (weights, neurons, channels) from a neural network to reduce model size and computational cost.

PUE (Power Usage Effectiveness)
A ratio that describes how efficiently a computer data centre uses energy; specifically, how much energy is used by the computing equipment (in contrast to cooling and other overhead).

Quantisation (AI Model)
The process of reducing the number of bits that represent a number; in AI, typically applied to model weights and activations to reduce model size and improve inference speed.

TPU (Tensor Processing Unit)
Google's custom-developed application-specific integrated circuit (ASIC) used to accelerate machine learning workloads.

Market Based (MB) Emissions
Refers to the calculation of greenhouse gas (GHG) emissions from purchased electricity, taking into account the specific energy contracts purchased by a company, rather than just the average emissions of the physical grid.

Training (AI)
The process of teaching an AI model to perform a specific task by feeding it large amounts of data.

Further exploration:
Resources for your sustainable AI journey

01 **Google Cloud's approach to responsible AI**

Our approach to developing and harnessing the potential of AI is grounded in our founding mission: to organise the world's information and make it universally accessible and useful.

Read more

02 **Google Cloud Sustainability**

Google Cloud Carbon Footprint tool and Region Picker tool

Read more

03 **Green Software Foundation**

SCI Specification and Green Software Patterns

Read more

04 **Google AI blog & research publications**

For latest on efficient models and responsible AI.

Read more

05 **White papers and reports**

The ROI of Gen AI

Read more

End note

a. Greenlight: Reductions in stops estimates are based on early data points from Google's analysis of traffic patterns before and after recommended adjustments to traffic signals that were implemented during tests conducted in 2024 and 2025. The reduction in stops can vary significantly, in some cases exceeding 30% for a period of time. Emissions reductions estimates are modeled using a Department of Energy emissions model. A single fuel based vehicle type is used as an approximation for all traffic, adjusted for country-level fleet mix from IEA data. These data points are averaged from coordinated intersections, and are subject to variation based on existing scenarios. We expect these estimates to evolve over time and look forward to sharing continued results as we perform additional analysis. To read more about Greenlight here.

b. Project Contrails: Using satellite imagery, large-scale weather data, and flight data, we trained a contrails prediction model. For this trial, we partnered with American Airlines to integrate contrail likely zone predictions into the tablets that their pilots used in flight so they could make real time adjustments in altitude to avoid creating contrails. We evaluated the model's performance using satellite imagery, comparing contrail formation (measured in contrail kilometers) in flights where pilots used predictions to avoid contrails, to contrail formation in flights where pilots didn't use contrail predictions. For more details, refer to the Project Contrails website and the Feasibility test of per-flight contrail avoidance in commercial aviation paper.

c. A point-in-time analysis quantified the energy consumed per median Gemini App text-generation prompt (0.24 watt-hours (Wh) of energy), considering data from May 2025. Emissions per prompt was estimated based on energy per prompt (0.03 grams of carbon dioxide equivalent (gCO_2e)), and applying Google's 2024 average fleetwide grid carbon intensity. Water consumption per prompt (0.26 milliliters (or about five drops) of water) was estimated based on energy per prompt, and applying Google's 2024 average fleetwide water usage effectiveness. These findings do not represent the specific environmental impact for all Gemini App text-generation prompts nor are they indicative of future performance. For more details, refer to the technical paper or summary in the Measuring the Environmental Impact of AI for Inference blog post.

End note

d. PUE is a standard industry ratio that compares the amount of non-computing overhead energy (used for things like cooling and power distribution) to the amount of energy used to power IT equipment (Google Environmental Report 2025). For example, a PUE of 2.0 means that for every watt of IT power, an additional watt is consumed to cool and distribute power to the IT equipment. A PUE closer to 1.0 means nearly all the energy is used for computing. According to the Uptime Institute's 2024 Global Data Center Survey, the global average PUE of respondents' data centers was 1.56. The overhead energy use comparison was calculated as follows: (1 - (Google's overhead energy use [0.09] divided by the industry average overhead energy use [0.56])) x 100 = 84%.

e. These calculations are based on internal data, as of March 2025. Google's TPU power efficiency relative to the earliest generation Cloud TPU v2 is measured by peak FP8 flops delivered per watt of thermal design power per chip package. For more details, refer to the technical paper.

f. A point-in-time analysis quantified the energy consumed per median Gemini App text-generation prompt, considering data from May 2025. These findings do not represent the specific environmental impact for all Gemini App text-generation prompts nor are they indicative of future performance. The results of the analysis from May 2025 were compared to baseline data from the median Gemini App text-generation prompt in May 2024. Energy per median prompt is subject to change as new models are added, AI model architecture evolves, and AI chatbot user behavior develops. The data and claims have not been verified by an independent third-party. The equivalency assumes a TV uses 100 Wh of energy per hour, per "How many watts does a TV use?," Energysage, November 2024, accessed July 2025. For more details, refer to the technical paper.

Acknowledgements

Contributors



John Abel

Managing Director
Office of the CTO, Google Cloud



Denise Pearl

Global Market Lead
Sustainability, Google Cloud



Jon Turnbull

Technical Program Manager
Environment x AI, Google Cloud



Michael Dietz

Customer Engineering Manager
Digital Natives, Google Cloud

Thank yous



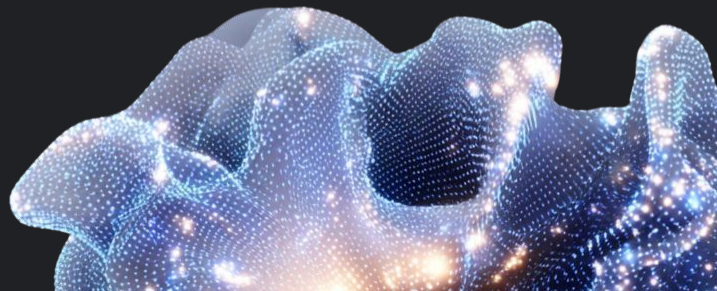
Sara Röste

Sustainability Marketing Lead
Google Cloud



Nivani Govinder

UKI Cloud Marketing
Google Cloud



Google Cloud



The new way to build software sustainably.

Let’s make it happen together.

Get in touch